

REQUIREMENTS DOCUMENT FOR THE VMU POWER-SPLIT MODE LOGIC

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1. INTRODUCTION

1.1 Adopted Conventions

To facilitate identification, allocation, and bidirectional traceability throughout this document, each requirement category uses a dedicated identifier convention:

- Use Cases are represented in the format [UC#];
- High-level system functional requirements are represented in the format [SysHLR#];
- High-level software functional requirements are represented in the format [SwHLR#];
- Low-level software functional requirements are represented in the format [SwLLR#];
- Non-functional requirements are represented in the format [NfHLR#];
- “#” represents the requirement number.

1.2 Problem Statement

Hybrid electric vehicles (HEVs) have become a key technology in the automotive industry's transition toward lower-emission and more energy-efficient mobility [1]. By combining an internal combustion engine with one or more electric machines and a traction battery, HEVs can recover braking energy, operate in pure-electric mode at low speeds, and reduce fuel consumption and CO₂ emissions when compared to conventional powertrains. This makes them especially relevant in regions with strict emission regulations and dense urban driving cycles, where short trips and frequent stops favor the use of the electric machine, while the engine remains available for higher-power demands and longer trips.

Among the available HEV architectures, the power-split topology is widely adopted because it allows the supervisory controller to flexibly distribute torque between the engine, the motor, and the generator according to the driving conditions. This flexibility, however, also requires a mode-management strategy that decides when the vehicle should operate as a pure electric vehicle, when the engine should be started, when regenerative braking is admissible, and when hybrid operation is the most appropriate choice. The correctness of this supervisory decision directly affects fuel economy, drivability, battery life, and the consistency of the powertrain behavior across drive cycles, which justifies treating the mode logic as a dedicated requirement subject in this document.

The HEV_powersplit_adapted model uses a power-split supervisory function called Control/Mode Logic. This model-level supervisor decides the operating mode and the enable signals required by the surrounding plant. The decision is based on vehicle speed, requested power, battery state of charge, and engine speed.

The Stateflow chart contains the top-level states StandStill, EV_mode, RegenB_mode, and Motion_mode_ICE, with the internal substates Start_mode, ICE_mode, and Hybrid_mode. For software implementation and traceability, this hierarchical behavior is represented in software

as six explicit operating modes corresponding to standstill, EV, regenerative braking, start, ICE, and hybrid behavior.

The engineering objective is to define a requirement baseline so that the power-split model and the future software implementation remain aligned to the same intended behavior. This document defines system requirements, software requirements, and non-functional requirements with explicit IDs so that vertical and horizontal traceability can follow from operational scenarios to model behavior and to the software that will be derived from it.

1.3 Purpose and Project Goals

1.3.1 Purpose

This document defines the requirements baseline for the VMU power-split mode logic used in the HEV_powersplit_adapted model and for the corresponding software implementation that will be derived from it. The baseline defines the intended system behavior, software behavior, and non-functional constraints for the VMU power-split mode logic. Figure 1 below illustrates the power-split hybrid drivetrain that is taken as the reference architecture in this document.

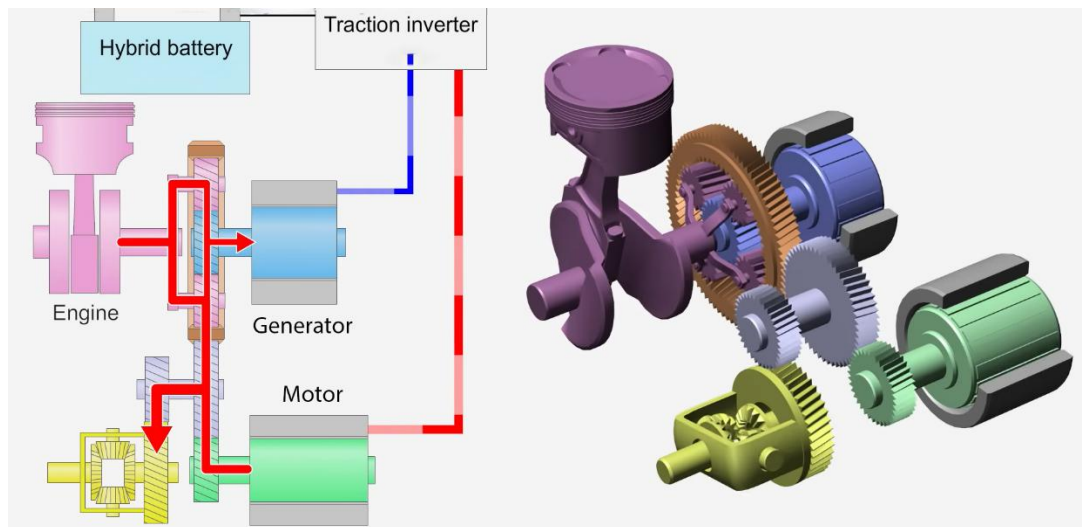


Figure 1: Power-split hybrid drivetrain model adopted as the reference architecture in this document. The internal combustion engine, the generator and the traction motor are linked through a planetary gear set, and electrical energy is exchanged with the battery through the traction inverter. Source: Authors.

1.3.2 Project Goals

The project goals for this baseline are to align the requirements with the current Stateflow chart, distinguish system, software, and non-functional requirements, and support traceability from use cases to model behavior and to the future software implementation.

- Reflect the actual states, transitions, inputs, outputs, and calibrations of the current power-split mode logic.
- Provide a clear split between system-level behavior, software behavior, and non-functional constraints.

Figure 2: Use case diagram of the VMU power-split supervisory mode logic, showing the central function Manage power-split operating mode, the operational use cases UC1–UC6 with their include and extend relationships, and the actors that interact with the system. Source: Authors.

Table 1. Summary of the VMU use cases

Identifier	Use case summary
UC1	Launch from standstill.
UC2	Low-speed EV propulsion.
UC3	Engine-supported propulsion when the EV window closes.
UC4	Braking-driven energy recovery.
UC5	Engine-supported mode reselection.
UC6	Return to standstill.

For this project, UC1 to UC6 serve as the stakeholder-level operational expectations for the Control/Mode Logic scope. Because no separate stakeholder requirement specification is maintained for this bounded supervisor, the system requirements in Section 4 trace back to these use cases as the originating stakeholder source.

2.1 Actors

Driver / drive-cycle demand source - In the current model, the driver request is represented indirectly by vehicle speed and requested power rather than by raw pedal signals. These signals define whether propulsion or regenerative operation is being requested from the mode logic.

Battery state-of-charge information - The battery-related interface provides the state-of-charge information used by the mode logic to admit or reject EV operation, regenerative operation, and entry into engine-supported propulsion.

Powertrain plant - The surrounding powersplit plant consumes Mot_Enable, Gen_Enable, and ICE_Enable and translates those supervisory enables into the detailed propulsion, engine, motor, and generator behavior of the complete model.

Calibration and verification engineer - This actor tunes the named thresholds, reviews requirement links, and prepares the later functional, unit-test, and static-analysis artifacts expected by the project deliveries.

2.2 Main supervisory function

The central supervisory use case is Manage power-split operation modes. At each 0.1 s cycle, the VMU evaluates the current operating point and resolves one valid mode request for the surrounding plant using vehicle speed, requested power, battery state of charge, and engine speed.

This central function also coordinates the deterministic enable outputs associated with the resolved mode. The detailed realization of that decision in StandStill, EV_mode, RegenB_mode, Start_mode, ICE_mode, and Hybrid_mode belongs to the state-machine view and therefore is specified later rather than expanded here as separate top-level use cases.

2.3 Operational use cases

UC1 - Launch from standstill is the entry point into active propulsion. Starting from StandStill, the supervisor decides whether the vehicle may enter EV propulsion or whether it must immediately enter the engine-supported branch because the EV window is not valid.

UC2 - Low-speed EV propulsion represents operation in EV_mode while vehicle speed, battery state of charge, and requested power remain inside the limits defined by the model. This use case remains conditional and is reevaluated at every cycle.

UC3 - Engine-supported propulsion when the EV window closes captures the point at which the EV window (the speed, requested-power and battery-SOC region in which EV operation is admissible) is no longer satisfied, so that the supervisor must enter the engine-supported branch of the powersplit logic. The EV window is considered closed when at least one of the following conditions occurs: vehicle speed leaves the EV speed band, requested power exceeds the EV power limit, or battery SOC drops below the EV exit threshold. In the current model, this branch is entered through Start_mode and may later be realized as ICE_mode or Hybrid_mode according to engine speed, requested power, and battery state of charge.

UC4 - Braking-driven energy recovery captures entry into RegenB_mode and its return paths. In this scenario, negative requested power and the current vehicle speed range determine whether the supervisor shall stay in regenerative operation, return to EV propulsion, return to standstill, or re-enter engine-supported propulsion.

UC5 - Engine-supported mode reselection covers the continued reevaluation that runs inside the engine-supported branch after entry through Start_mode. This use case captures both the internal powersplit supervision (Start, ICE and Hybrid sub-states reselecting each other according to engine speed, requested power and battery state of charge) and the external exits of Motion_mode_ICE back to the regenerative-braking mode, the EV mode and the standstill mode. External exits are evaluated before internal transitions, so any admissible return path leaves the engine-supported branch instead of triggering an internal reselection.

UC6 - Return to standstill closes the cycle by bringing the supervisor back to the base state when vehicle speed falls below the stop threshold and requested power remains inside the neutral stop window, including returns from EV_mode, RegenB_mode, and the engine-supported branch.

2.4 Boundary conditions and out-of-scope behavior

The current Stateflow chart does not include a degraded state, a fault manager, friction-brake allocation logic, a detailed torque-split controller, or dedicated supervisory use cases for low-gear, high-gear, and top-gear powersplit conditions. Those behaviors are outside the Control/Mode Logic scope or are implemented in other parts of the surrounding plant model.

For that reason, this requirement document focuses only on the behavior actually present in Control/Mode Logic: selection of one valid supervisory mode, hysteretic transition criteria, request of engine-supported or regenerative operation when needed, and deterministic enable outputs. If degraded behavior is added later, it should enter the document as a separate requirement branch with its own explicit requirements and traceability.

2.5 Overall interpretation of the diagram

Taken as a whole, the powersplit use case view presents the VMU as a supervisory function that interprets the current operating point and coordinates the corresponding plant-level mode request. Its main role is to keep launch, EV propulsion, engine-supported propulsion,

regenerative operation, engine-supported reselection, and return to standstill mutually consistent with the plant-facing enable outputs.

From a system-engineering perspective, this use case view stays at the stakeholder-facing level while still reflecting the powersplit-specific behavior of the implemented model. It therefore provides a cleaner source for the system requirements, while the finer distinctions between Start_mode, ICE_mode, and Hybrid_mode remain in the state-machine and software requirement sections where they belong.

3. STATE DIAGRAM

Figure 3 represents the state diagram implemented in Control/Mode Logic. It defines the supervisory behavior of the current HEV_powersplit_adapted model and executes with the discrete sample time HEV_Param.Control.Mode_Logic_TS = 0.1 s. At the top level, the chart contains StandStill, EV_mode, RegenB_mode, and Motion_mode_ICE. Motion_mode_ICE contains the internal states Start_mode, ICE_mode, and Hybrid_mode, with entry at Start_mode.

The chart outputs are supervisory enable requests associated with the active state: StandStill -> (0,0,0); EV_mode -> (1,0,0); RegenB_mode -> (1,0,0); Start_mode -> (1,1,0); ICE_mode -> (0,1,1); Hybrid_mode -> (1,1,1). These outputs are consumed elsewhere in the plant and therefore shall be interpreted as deterministic mode-dependent requests.

The transition guards evaluate vehicle speed [km/h], requested power [kW], battery state of charge [-], and engine speed [rpm] against the named calibration constants summarized below. The compact aliases used in the transition tables correspond to the model signals speed, P_dem, SOC/charge, and wEng/engine_speed. The numerical values used for these calibrations were chosen by the project team to match the operating envelope of the source model; the upper bound of the EV speed window (SPEED_EV_MAX) is supported by reference [1] in Section 8.

Constant	Value	Role in transitions
ENG_ON	800 rpm	Engine-on threshold used by Motion_mode_ICE exits and by Start_mode transitions to ICE_mode or Hybrid_mode.
ENG_OFF	790 rpm	Engine-off threshold used by ICE_mode and Hybrid_mode returns to Start_mode.
SPEED_STOP	0.5 km/h	Standstill threshold used by returns to StandStill.
SPEED_REGEN	5 km/h	Minimum speed threshold used for entry into RegenB_mode.
SPEED_EV_MAX	35 km/h	Upper speed limit of the EV operating window and a boundary for engine-supported motion [1].
PDEM_REGEN	-5 kW	Requested-power threshold that admits regenerative braking operation.
PDEM_STOP_LOW	-1 kW	Lower bound of the neutral stop-demand window used by returns to StandStill.
PDEM_STOP_HIGH	1 kW	Upper bound of the neutral stop-demand window used by returns to StandStill.
PDEM_HYB_IN	50 kW	Requested-power threshold that forces EV_mode to leave for Motion_mode_ICE.
PDEM_HYB_OUT	40 kW	Requested-power limit that allows Motion_mode_ICE to return to EV_mode.

below use the compact model aliases speed [km/h], P_dem [kW], SOC = battery state of charge [-], and wEng = engine speed [rpm].

3.1.1 Top-Level Transitions

ID	Source State	Destination State	Transition Condition
T1	StandStill	EV_mode	speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_IN
T2	StandStill	Motion_mode_ICE	speed > SPEED_STOP && (speed > SPEED_EV_MAX SOC < SOC_EV_IN)
T3	EV_mode	RegenB_mode	speed > SPEED_REGEN && P_dem <= PDEM_REGEN
T4	EV_mode	Motion_mode_ICE	speed > SPEED_EV_MAX P_dem >= PDEM_HYB_IN SOC < SOC_EV_OUT
T5	EV_mode	StandStill	speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW
T6	RegenB_mode	Motion_mode_ICE	((speed > SPEED_EV_MAX) && (P_dem >= PDEM_STOP_LOW)) SOC < SOC_EV_OUT
T7	RegenB_mode	StandStill	speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW
T8	RegenB_mode	EV_mode	P_dem >= PDEM_STOP_LOW && speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_OUT
T9	Motion_mode_ICE	RegenB_mode	wEng > ENG_ON && speed > SPEED_REGEN && P_dem <= PDEM_REGEN
T10	Motion_mode_ICE	EV_mode	wEng > ENG_ON && P_dem <= PDEM_HYB_OUT && P_dem >= PDEM_STOP_LOW && speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_IN
T11	Motion_mode_ICE	StandStill	speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW

3.1.2 Internal Transitions of Motion_mode_ICE

ID	Source State	Destination State	Transition Condition
T12	Start_mode	Hybrid_mode	wEng > ENG_ON && SOC >= SOC_MID && (speed > SPEED_EV_MAX P_dem >= PDEM_HYB_MID)

ID	Source State	Destination State	Transition Condition
T13	Start_mode	ICE_mode	wEng > ENG_ON && (SOC < SOC_MID (speed <= SPEED_EV_MAX && P_dem < PDEM_HYB_MID))
T14	ICE_mode	Start_mode	wEng <= ENG_OFF
T15	ICE_mode	Hybrid_mode	P_dem >= PDEM_HYB_MID && SOC >= SOC_MID
T16	Hybrid_mode	Start_mode	wEng <= ENG_OFF
T17	Hybrid_mode	ICE_mode	P_dem <= PDEM_HYB_LOW SOC < SOC_LOW

4. FUNCTIONAL REQUIREMENTS

This section defines the functional requirement baseline for the VMU power-split mode logic. The system requirements preserve the supervisory intent at vehicle-function level, the high-level software requirements organize the software behavior by state family, and the low-level software requirements capture each executable transition individually.

4.1 High-level system functional requirements

4.1.1 [SysHLR01] Power-Split Supervisory Mode Selection

Requirement statement (shall)	The VMU shall select exactly one operating mode from {standstill mode, EV mode, regenerative-braking mode, Start mode, ICE mode, and Hybrid mode} at each mode-logic execution cycle by evaluating vehicle speed [km/h], requested power [kW], battery state of charge [-], and engine speed [rpm].
Conditions / Applicability	Active whenever Control/Mode Logic is executed in nominal mode-management operation.
Stakeholder Priority	High
Rationale	Establishes the single-state supervisory decision expected by the model and the software.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

4.1.2 [SysHLR02] Deterministic Powertrain Enable Coordination

Requirement statement (shall)	For each selected operating mode, the VMU shall command deterministic motor, generator, and engine enable requests according to the mode-dependent pattern defined by the model: standstill {0,0,0}, EV {1,0,0}, regenerative braking {1,0,0}, Start {1,1,0}, ICE {0,1,1}, and Hybrid {1,1,1}.
Conditions / Applicability	Applies in each healthy mode-logic cycle after the next mode has been resolved.
Stakeholder Priority	High
Rationale	Ensures supervisory mode selection and plant-facing enable outputs remain consistent.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

4.1.3 [SysHLR03] Calibratable and Hysteretic Transition Strategy

Requirement statement (shall)	The VMU shall evaluate mode transitions using the named calibrations ENG_ON, ENG_OFF, SPEED_STOP, SPEED_REGEN, SPEED_EV_MAX, PDEM_REGEN, PDEM_STOP_LOW, PDEM_STOP_HIGH, PDEM_HYB_IN, PDEM_HYB_OUT, PDEM_HYB_MID, PDEM_HYB_LOW, SOC_EV_IN, SOC_EV_OUT, SOC_MID, and SOC_LOW while preserving the modeled hysteresis on battery state of charge and engine speed thresholds.
Conditions / Applicability	Applies to every transition decision in the mode logic.
Stakeholder Priority	High
Rationale	Keeps the requirement baseline aligned with the calibrated model and preserves the intended hysteresis behavior.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

4.2 High-level software functional requirements

The high-level software requirements below organize the software representation by initialization, output mapping, state-family decision logic, and software-defined behaviors that remain outside the modeled transition set. In this section, the software modes correspond to standstill, EV, regenerative braking, start, ICE, and hybrid behavior represented by the model. These requirements act as parents for the transition-level low-level software requirements defined in Section 4.3.

4.2.1 [SwHLR01] Initialization Behavior

Requirement statement (shall)	The software shall initialize the active software mode to the standstill mode before the first normal mode-evaluation cycle.
Conditions / Applicability	Applies during software initialization and before the first periodic mode-evaluation cycle.
Derived from	SysHLR01
Stakeholder Priority	High
Rationale	Creates a deterministic starting point for the software representation of the state machine.
Related UC(s)	UC1, UC6

4.2.2 [SwHLR02] Deterministic Output Mapping by Active Mode

Requirement statement (shall)	The software shall command the enable outputs associated with the resolved active software mode according to the mode-to-output mapping of the source model.
Conditions / Applicability	Applies after each mode-resolution decision.
Derived from	SysHLR02
Stakeholder Priority	High
Rationale	Separates output assignment from transition selection and avoids duplicating output obligations in every low-level transition requirement.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

4.2.3 [SwHLR03] Standstill Mode Transition Family

Requirement statement (shall)	While the active software mode is the standstill mode, the software shall evaluate exits to EV mode and Start mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode represents standstill behavior.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the standstill decision family once and allocates the detailed guards to derived low-level requirements.
Related UC(s)	UC1

4.2.4 [SwHLR04] EV Mode Transition Family

Requirement statement (shall)	While the active software mode is EV mode, the software shall evaluate exits to regenerative-braking mode, Start mode, and standstill mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode represents EV behavior.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the EV decision family once and allocates the detailed guards to derived low-level requirements.
Related UC(s)	UC2, UC3, UC4, UC6

4.2.5 [SwHLR05] Regenerative-Braking Transition Family

Requirement statement (shall)	While the active software mode is the regenerative-braking mode, the software shall evaluate exits to Start mode, standstill mode, and EV mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode represents regenerative-braking behavior.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the regenerative-braking decision family once and allocates the detailed guards to derived low-level requirements.
Related UC(s)	UC2, UC3, UC4, UC6

4.2.6 [SwHLR06] Engine-Supported External Exit Family

Requirement statement (shall)	While the active software mode is Start, ICE, or Hybrid, the software shall evaluate exits to regenerative-braking mode, EV mode, and standstill mode before any internal transition within the engine-supported mode representation.
Conditions / Applicability	Applies while the active software mode represents Start, ICE, or Hybrid behavior.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High

Rationale	Captures the common external-exit policy of Motion_mode_ICE and provides the parent requirement for the nine derived low-level exits.
Related UC(s)	UC2, UC4, UC6

4.2.7 [SwHLR07] Start Mode Internal Transition Family

Requirement statement (shall)	While the active software mode is Start and no higher-priority engine-supported external exit is active, the software shall evaluate internal transitions to Hybrid mode and ICE mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode is Start and the external-exit family does not trigger.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the internal Start_mode decision family and provides the parent requirement for the derived low-level transitions.
Related UC(s)	UC3, UC5

4.2.8 [SwHLR08] ICE Mode Internal Transition Family

Requirement statement (shall)	While the active software mode is ICE and no higher-priority engine-supported external exit is active, the software shall evaluate internal transitions to Start mode and Hybrid mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode is ICE and the external-exit family does not trigger.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the internal ICE_mode decision family and provides the parent requirement for the derived low-level transitions.
Related UC(s)	UC3, UC5

4.2.9 [SwHLR09] Hybrid Mode Internal Transition Family

Requirement statement (shall)	While the active software mode is Hybrid and no higher-priority engine-supported external exit is active, the software shall evaluate internal transitions to Start mode and ICE mode in the priority order defined by the model.
Conditions / Applicability	Applies while the active software mode is Hybrid and the external-exit family does not trigger.
Derived from	SysHLR01, SysHLR03
Stakeholder Priority	High
Rationale	Captures the internal Hybrid_mode decision family and provides the parent requirement for the derived low-level transitions.
Related UC(s)	UC3, UC5

4.2.10 [SwHLR10] Initialization and Invalid-Mode Recovery Outside the Modeled Transition Set

Requirement statement (shall)	The software shall treat initialization to the standstill mode and recovery from an invalid active-mode value as software-defined behaviors outside the 23 modeled transitions.
Conditions / Applicability	Applies during initialization and whenever the stored active mode does not match a valid software mode.
Derived from	SysHLR01
Stakeholder Priority	High
Rationale	Separates startup and defensive recovery behavior from the modeled transition set while keeping the implementation behavior explicit in the requirement baseline.
Related UC(s)	UC1, UC6

4.3 Low-level software functional requirements

The low-level software requirements below provide one requirement per executable transition. The three chart-level exits from Motion_mode_ICE are allocated in software as one requirement per active software mode in the start, ICE, and hybrid branches, so the low-level section contains 23 transition requirements. This count excludes initialization to the standstill mode and recovery from an invalid active-mode value, because those behaviors are software-defined and are not modeled transitions.

4.3.1 [SwLLR01] Standstill Mode to EV Mode

Requirement statement (shall)	When the active software mode is the standstill mode and speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_IN, the software shall transition to EV mode.
Conditions / Applicability	Applies while the active software mode is the standstill mode. This is the highest-priority matching exit.
Derived from	SwHLR02, SwHLR03
Stakeholder Priority	High
Rationale	Implements the admissible EV launch path from standstill.
Related UC(s)	UC1, UC2

4.3.2 [SwLLR02] Standstill Mode to Start Mode

Requirement statement (shall)	When the active software mode is the standstill mode and speed > SPEED_STOP && (speed > SPEED_EV_MAX SOC < SOC_EV_IN), the software shall transition to Start mode.
Conditions / Applicability	Applies while the active software mode is the standstill mode, after the higher-priority EV exit has been excluded. Entry into Motion_mode_ICE is represented in software by Start mode.
Derived from	SwHLR02, SwHLR03
Stakeholder Priority	High
Rationale	Implements the direct launch path into engine-supported propulsion when the EV window is not valid.
Related UC(s)	UC1, UC3

4.3.3 [SwLLR03] EV Mode to Regenerative-Braking Mode

Requirement statement (shall)	When the active software mode is EV mode and speed > SPEED_REGEN && P_dem <= PDEM_REGEN, the software shall transition to the regenerative-braking mode.
Conditions / Applicability	Applies while the active software mode is EV mode. This is the highest-priority EV exit.
Derived from	SwHLR02, SwHLR04
Stakeholder Priority	High
Rationale	Implements entry into regenerative braking from EV operation.
Related UC(s)	UC4

4.3.4 [SwLLR04] EV Mode to Start Mode

Requirement statement (shall)	When the active software mode is EV mode and (speed > SPEED_EV_MAX P_dem >= PDEM_HYB_IN SOC < SOC_EV_OUT), the software shall transition to Start mode.
Conditions / Applicability	Applies while the active software mode is EV mode, after the higher-priority regenerative-braking exit has been excluded. Entry into Motion_mode_ICE is represented in software by Start mode.
Derived from	SwHLR02, SwHLR04
Stakeholder Priority	High
Rationale	Implements the EV exit to engine-supported propulsion when the EV window closes or demand increases.
Related UC(s)	UC3

4.3.5 [SwLLR05] EV Mode to Standstill Mode

Requirement statement (shall)	When the active software mode is EV mode and speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW, the software shall transition to the standstill mode.
Conditions / Applicability	Applies while the active software mode is EV mode, after the higher-priority regenerative-braking and engine-supported exits have been excluded.
Derived from	SwHLR02, SwHLR04
Stakeholder Priority	High
Rationale	Implements the EV return to standstill in the neutral stop-demand window.
Related UC(s)	UC6

4.3.6 [SwLLR06] Regenerative-Braking Mode to Start Mode

Requirement statement (shall)	When the active software mode is the regenerative-braking mode and (((speed > SPEED_EV_MAX) && (P_dem >= PDEM_STOP_LOW)) SOC < SOC_EV_OUT), the software shall transition to Start mode.
Conditions / Applicability	Applies while the active software mode is the regenerative-braking mode. This is the highest-priority regenerative-braking

	exit. Entry into Motion_mode_ICE is represented in software by Start mode.
Derived from	SwHLR02, SwHLR05
Stakeholder Priority	High
Rationale	Implements the return from regenerative braking to engine-supported propulsion when the EV window is no longer valid.
Related UC(s)	UC3, UC4

4.3.7 [SwLLR07] Regenerative-Braking Mode to Standstill Mode

Requirement statement (shall)	When the active software mode is the regenerative-braking mode and speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW, the software shall transition to the standstill mode.
Conditions / Applicability	Applies while the active software mode is the regenerative-braking mode, after the higher-priority engine-supported exit has been excluded.
Derived from	SwHLR02, SwHLR05
Stakeholder Priority	High
Rationale	Implements braking completion into standstill from the regenerative-braking mode.
Related UC(s)	UC6

4.3.8 [SwLLR08] Regenerative-Braking Mode to EV Mode

Requirement statement (shall)	When the active software mode is the regenerative-braking mode and P_dem >= PDEM_STOP_LOW && speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_OUT, the software shall transition to EV mode.
Conditions / Applicability	Applies while the active software mode is the regenerative-braking mode, after the higher-priority engine-supported and standstill exits have been excluded.
Derived from	SwHLR02, SwHLR05
Stakeholder Priority	High
Rationale	Implements the return from regenerative braking to EV propulsion.
Related UC(s)	UC2, UC4

4.3.9 [SwLLR09] Start Mode to Regenerative-Braking Mode

Requirement statement (shall)	When the active software mode is Start mode and wEng > ENG_ON && speed > SPEED_REGEN && P_dem <= PDEM_REGEN, the software shall transition to the regenerative-braking mode.
Conditions / Applicability	Applies while the active software mode is Start mode. This is the highest-priority external exit from the engine-supported mode representation.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High

Rationale	Allocates the chart-level Motion_mode_ICE to RegenB exit to the Start-mode software branch.
Related UC(s)	UC4, UC5

4.3.10 [SwLLR10] ICE Mode to Regenerative-Braking Mode

Requirement statement (shall)	When the active software mode is ICE mode and wEng > ENG_ON && speed > SPEED_REGEN && P_dem <= PDEM_REGEN, the software shall transition to the regenerative-braking mode.
Conditions / Applicability	Applies while the active software mode is ICE mode. This is the highest-priority external exit from ICE mode.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to RegenB exit to the ICE-mode software branch.
Related UC(s)	UC4, UC5

4.3.11 [SwLLR11] Hybrid Mode to Regenerative-Braking Mode

Requirement statement (shall)	When the active software mode is Hybrid mode and wEng > ENG_ON && speed > SPEED_REGEN && P_dem <= PDEM_REGEN, the software shall transition to the regenerative-braking mode.
Conditions / Applicability	Applies while the active software mode is Hybrid mode. This is the highest-priority external exit from Hybrid mode.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to RegenB exit to the Hybrid-mode software branch.
Related UC(s)	UC4, UC5

4.3.12 [SwLLR12] Start Mode to EV Mode

Requirement statement (shall)	When the active software mode is Start mode and wEng > ENG_ON && P_dem <= PDEM_HYB_OUT && P_dem >= PDEM_STOP_LOW && speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_IN, the software shall transition to EV mode.
Conditions / Applicability	Applies while the active software mode is Start mode, after the higher-priority regenerative-braking exit has been excluded. External exits are evaluated before internal Start-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to EV exit to the Start-mode software branch.
Related UC(s)	UC2, UC5

4.3.13 [SwLLR13] ICE Mode to EV Mode

Requirement statement (shall)	When the active software mode is ICE mode and wEng > ENG_ON && P_dem <= PDEM_HYB_OUT && P_dem >= PDEM_STOP_LOW && speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_IN, the software shall transition to EV mode.
Conditions / Applicability	Applies while the active software mode is ICE mode, after the higher-priority regenerative-braking exit has been excluded. External exits are evaluated before internal ICE-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to EV exit to the ICE-mode software branch.
Related UC(s)	UC2, UC5

4.3.14 [SwLLR14] Hybrid Mode to EV Mode

Requirement statement (shall)	When the active software mode is Hybrid mode and wEng > ENG_ON && P_dem <= PDEM_HYB_OUT && P_dem >= PDEM_STOP_LOW && speed > SPEED_STOP && speed <= SPEED_EV_MAX && SOC >= SOC_EV_IN, the software shall transition to EV mode.
Conditions / Applicability	Applies while the active software mode is Hybrid mode, after the higher-priority regenerative-braking exit has been excluded. External exits are evaluated before internal Hybrid-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to EV exit to the Hybrid-mode software branch.
Related UC(s)	UC2, UC5

4.3.15 [SwLLR15] Start Mode to Standstill Mode

Requirement statement (shall)	When the active software mode is Start mode and speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW, the software shall transition to the standstill mode.
Conditions / Applicability	Applies while the active software mode is Start mode, after the higher-priority regenerative-braking and EV exits have been excluded. This is the third external exit checked before internal Start-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to StandStill exit to the Start-mode software branch.
Related UC(s)	UC5, UC6

4.3.16 [SwLLR16] ICE Mode to Standstill Mode

Requirement statement (shall)	When the active software mode is ICE mode and speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW, the software shall transition to the standstill mode.
Conditions / Applicability	Applies while the active software mode is ICE mode, after the higher-priority regenerative-braking and EV exits have been excluded. This is the third external exit checked before the remaining internal ICE-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to StandStill exit to the ICE-mode software branch.
Related UC(s)	UC5, UC6

4.3.17 [SwLLR17] Hybrid Mode to Standstill Mode

Requirement statement (shall)	When the active software mode is Hybrid mode and speed <= SPEED_STOP && P_dem <= PDEM_STOP_HIGH && P_dem >= PDEM_STOP_LOW, the software shall transition to the standstill mode.
Conditions / Applicability	Applies while the active software mode is Hybrid mode, after the higher-priority regenerative-braking and EV exits have been excluded. This is the third external exit checked before the remaining internal Hybrid-mode transitions.
Derived from	SwHLR02, SwHLR06
Stakeholder Priority	High
Rationale	Allocates the chart-level Motion_mode_ICE to StandStill exit to the Hybrid-mode software branch.
Related UC(s)	UC5, UC6

4.3.18 [SwLLR18] Start Mode to Hybrid Mode

Requirement statement (shall)	When the active software mode is Start mode and wEng > ENG_ON && SOC >= SOC_MID && (speed > SPEED_EV_MAX P_dem >= PDEM_HYB_MID), the software shall transition to Hybrid mode.
Conditions / Applicability	Applies while the active software mode is Start mode, after all higher-priority external exits have been excluded. This is the highest-priority remaining internal Start-mode transition.
Derived from	SwHLR02, SwHLR07
Stakeholder Priority	High
Rationale	Implements the internal Start_mode transition to Hybrid mode.
Related UC(s)	UC3, UC5

4.3.19 [SwLLR19] Start Mode to ICE Mode

Requirement statement (shall)	When the active software mode is Start mode and wEng > ENG_ON && (SOC < SOC_MID (speed <= SPEED_EV_MAX
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	&& P_dem < PDEM_HYB_MID)), the software shall transition to ICE mode.
Conditions / Applicability	Applies while the active software mode is Start mode, after all higher-priority external exits and the higher-priority internal Hybrid-mode transition have been excluded.
Derived from	SwHLR02, SwHLR07
Stakeholder Priority	High
Rationale	Implements the internal Start_mode transition to ICE mode.
Related UC(s)	UC3, UC5

4.3.20 [SwLLR20] ICE Mode to Start Mode

Requirement statement (shall)	When the active software mode is ICE mode and wEng <= ENG_OFF, the software shall transition to Start mode.
Conditions / Applicability	Applies while the active software mode is ICE mode, after all higher-priority external exits have been excluded. This is the highest-priority remaining internal ICE-mode transition.
Derived from	SwHLR02, SwHLR08
Stakeholder Priority	High
Rationale	Implements the engine-off return from ICE mode to Start mode.
Related UC(s)	UC3, UC5

4.3.21 [SwLLR21] ICE Mode to Hybrid Mode

Requirement statement (shall)	When the active software mode is ICE mode and P_dem >= PDEM_HYB_MID && SOC >= SOC_MID, the software shall transition to Hybrid mode.
Conditions / Applicability	Applies while the active software mode is ICE mode, after all higher-priority external exits and the higher-priority internal Start-mode return have been excluded.
Derived from	SwHLR02, SwHLR08
Stakeholder Priority	High
Rationale	Implements the internal ICE_mode transition to Hybrid mode.
Related UC(s)	UC3, UC5

4.3.22 [SwLLR22] Hybrid Mode to Start Mode

Requirement statement (shall)	When the active software mode is Hybrid mode and wEng <= ENG_OFF, the software shall transition to Start mode.
Conditions / Applicability	Applies while the active software mode is Hybrid mode, after all higher-priority external exits have been excluded. This is the highest-priority internal transition from Hybrid mode.
Derived from	SwHLR02, SwHLR09
Stakeholder Priority	High
Rationale	Implements the internal Hybrid_mode return to Start mode when engine speed falls to the engine-off threshold.
Related UC(s)	UC3, UC5

4.3.23 [SwLLR23] Hybrid Mode to ICE Mode

Requirement statement (shall)	When the active software mode is Hybrid mode and ($P_dem \leq PDEM_HYB_LOW \parallel SOC < SOC_LOW$), the software shall transition to ICE mode.
Conditions / Applicability	Applies while the active software mode is Hybrid mode, after all higher-priority external exits and the higher-priority internal Start-mode return have been excluded.
Derived from	SwHLR02, SwHLR09
Stakeholder Priority	High
Rationale	Implements the internal Hybrid_mode transition back to ICE mode.
Related UC(s)	UC3, UC5

5. NON-FUNCTIONAL REQUIREMENTS

This section defines non-functional constraints that apply to the same mode-logic baseline. These requirements complement Section 4 by constraining timing, stability, resource usage, and deterministic behavior of the implementation.

5.1 [NfHLR01] Periodic Execution and Sample Time

Requirement statement (shall)	The software shall execute at a fixed periodic rate of 0.1 s consistent with the mode-logic sample-time calibration and resolve the next mode within the same sample period.
Conditions / Applicability	Applies to every periodic execution of the mode-logic software during nominal operation.
Stakeholder Priority	High
Rationale	Preserves the timing semantics of the source model and prevents backlog between control cycles.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

5.2 [NfHLR02] Transition Stability and Hysteresis Preservation

Requirement statement (shall)	The software shall preserve the modeled hysteresis at the battery-state-of-charge boundary (entry at SOC_EV_IN, exit at SOC_EV_OUT) and at the engine-speed boundary (engine on at ENG_ON, engine off at ENG_OFF) while evaluating source-state exits in deterministic priority order.
Conditions / Applicability	Applies to all mode-transition evaluations, especially near calibrated boundaries.
Stakeholder Priority	High
Rationale	Prevents oscillation near thresholds and keeps the software behavior aligned with the Stateflow chart.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

5.3 [NfHLR03] Static Allocation and Bounded State Storage

Requirement statement (shall)	The mode-logic implementation shall use only fixed-size pre-allocated storage, without dynamic memory allocation, recursion, or variable-length storage.
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Conditions / Applicability	Applies to the mode-logic software components and the associated runtime data structures.
Stakeholder Priority	High
Rationale	Supports predictable resource usage and straightforward embedded deployment.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

5.4 [NfHLR04] Deterministic Outputs and Reproducibility

Requirement statement (shall)	Given the same initial mode and the same input sequence, the software shall produce the same sequence of operating modes and enable outputs across repeated executions of the same release baseline.
Conditions / Applicability	Applies to repeated runs of the same software baseline under the same nominal build configuration.
Stakeholder Priority	High
Rationale	Supports repeatable analysis, debugging, and regression comparison.
Related UC(s)	UC1, UC2, UC3, UC4, UC5, UC6

6. TRACEABILITY

The traceability tables below connect the use cases, the system requirements, the software requirement hierarchy, and the Stateflow transition basis that supports the supervisory behavior. This keeps the document organized now while leaving space for a later coverage table that will link these requirements to the future software implementation.

6.1 Traceability Between Stakeholder Use Cases and System Requirements

Use Cases / System requirements	SysHLR01	SysHLR02	SysHLR03
UC1	X	X	X
UC2	X	X	X
UC3	X	X	X
UC4	X	X	X
UC5	X	X	X
UC6	X	X	X

6.2 Traceability Between System Requirements and High-Level Software Requirements

System requirement	Supporting high-level software requirement(s)
SysHLR01	SwHLR01, SwHLR03, SwHLR04, SwHLR05, SwHLR06, SwHLR07, SwHLR08, SwHLR09, SwHLR10
SysHLR02	SwHLR02
SysHLR03	SwHLR03, SwHLR04, SwHLR05, SwHLR06, SwHLR07, SwHLR08, SwHLR09

6.3 Traceability Between High-Level and Low-Level Software Requirements

High-level software requirement	Derived low-level software requirement(s)
SwHLR01	No derived transition requirement; covered directly by the initialization behavior.
SwHLR02	Supports SwLLR01 to SwLLR23 by defining the mode-to-output mapping used after each transition.
SwHLR03	SwLLR01 to SwLLR02
SwHLR04	SwLLR03 to SwLLR05
SwHLR05	SwLLR06 to SwLLR08
SwHLR06	SwLLR09 to SwLLR17
SwHLR07	SwLLR18 to SwLLR19
SwHLR08	SwLLR20 to SwLLR21
SwHLR09	SwLLR22 to SwLLR23
SwHLR10	No derived low-level transition requirement; covers initialization to the standstill mode and invalid-mode recovery outside the 23 modeled transitions.

6.4 Traceability Between Software Requirements and Stateflow Basis

Software requirement	Stateflow basis
SwHLR01	Chart entry to StandStill
SwHLR02	Mode-dependent output mapping
SwHLR03	StandStill exit family
SwHLR04	EV exit family
SwHLR05	RegenB exit family
SwHLR06	Motion_mode_ICE external exits
SwHLR07	Start_mode internal exits
SwHLR08	ICE_mode internal exits
SwHLR09	Hybrid_mode internal exits
SwHLR10	Software-defined behavior outside the modeled transition set
SwLLR01	T1: StandStill -> EV_mode
SwLLR02	T2: StandStill -> Motion_mode_ICE/Start_mode
SwLLR03	T3: EV_mode -> RegenB_mode
SwLLR04	T4: EV_mode -> Motion_mode_ICE/Start_mode
SwLLR05	T5: EV_mode -> StandStill
SwLLR06	T6: RegenB_mode -> Motion_mode_ICE/Start_mode
SwLLR07	T7: RegenB_mode -> StandStill
SwLLR08	T8: RegenB_mode -> EV_mode
SwLLR09	Derived from T9: Motion_mode_ICE/Start_mode -> RegenB_mode

Software requirement	Stateflow basis
SwLLR10	Derived from T9: Motion_mode_ICE/ICE_mode -> RegenB_mode
SwLLR11	Derived from T9: Motion_mode_ICE/Hybrid_mode -> RegenB_mode
SwLLR12	Derived from T10: Motion_mode_ICE/Start_mode -> EV_mode
SwLLR13	Derived from T10: Motion_mode_ICE/ICE_mode -> EV_mode
SwLLR14	Derived from T10: Motion_mode_ICE/Hybrid_mode -> EV_mode
SwLLR15	Derived from T11: Motion_mode_ICE/Start_mode -> StandStill
SwLLR16	Derived from T11: Motion_mode_ICE/ICE_mode -> StandStill
SwLLR17	Derived from T11: Motion_mode_ICE/Hybrid_mode -> StandStill
SwLLR18	T12: Start_mode -> Hybrid_mode
SwLLR19	T13: Start_mode -> ICE_mode
SwLLR20	T14: ICE_mode -> Start_mode
SwLLR21	T15: ICE_mode -> Hybrid_mode
SwLLR22	T16: Hybrid_mode -> Start_mode
SwLLR23	T17: Hybrid_mode -> ICE_mode

7. CONCLUSION

This document establishes the requirements baseline for the VMU power-split mode logic used in the HEV_powersplit_adapted model. It defines the expected supervisory behavior for mode selection, transition evaluation, and mode-dependent enable outputs, while keeping the specification aligned with the reference Stateflow chart.

The resulting requirement set provides a traceable basis for the next development activities, including software implementation, model-to-code alignment, unit testing, and functional verification. By linking use cases, system requirements, software requirements, and Stateflow transition behavior, the document supports consistent development within the defined project scope.

8. REFERENCES

- [1] U.S. DEPARTMENT OF ENERGY; U.S. ENVIRONMENTAL PROTECTION AGENCY. **Hybrid technology**. fueleconomy.gov, [n.d.]. Available at: <https://www.fueleconomy.gov/feg/hybridtech.shtml>. Accessed on: 3 May 2026.

Appendix A - Glossary

Term	Definition
Vehicle speed (speed)	Vehicle-speed input used by Control/Mode Logic to detect standstill, EV eligibility, regenerative entry, and exits from engine-supported operation.
Requested power (P_dem)	Requested-power input used to distinguish propulsion demand, regenerative demand, neutral stop demand, and hybrid-entry thresholds.
Battery state of charge (SOC / charge)	Battery state-of-charge input used to admit or reject EV operation and to select ICE or Hybrid behavior inside Motion_mode_ICE.
Engine speed (wEng / engine_speed)	Engine-speed input used for engine-on and engine-off hysteresis in Motion_mode_ICE.
StandStill	Top-level Stateflow state representing the stationary mode with Mot_Enable = 0, Gen_Enable = 0, and ICE_Enable = 0.
EV_mode	Top-level Stateflow state representing electric propulsion availability with Mot_Enable = 1, Gen_Enable = 0, and ICE_Enable = 0.
RegenB_mode	Top-level Stateflow state representing regenerative braking behavior with Mot_Enable = 1, Gen_Enable = 0, and ICE_Enable = 0.
Motion_mode_ICE	Top-level grouped Stateflow state for engine-supported operation containing Start_mode, ICE_mode, and Hybrid_mode.
Start_mode	Internal Motion_mode_ICE state entered before ICE_mode or Hybrid_mode, with Mot_Enable = 1, Gen_Enable = 1, and ICE_Enable = 0.

ICE_mode	Internal Motion_mode_ICE state for engine-supported operation with Mot_Enable = 0, Gen_Enable = 1, and ICE_Enable = 1.
Hybrid_mode	Internal Motion_mode_ICE state for hybrid operation with Mot_Enable = 1, Gen_Enable = 1, and ICE_Enable = 1.
Standstill software mode	Software-mode representation corresponding to StandStill in the model.
EV software mode	Software-mode representation corresponding to EV_mode in the model.
Regenerative-braking software mode	Software-mode representation corresponding to RegenB_mode in the model.
Start software mode	Software-mode representation corresponding to Motion_mode_ICE/Start_mode in the model.
ICE software mode	Software-mode representation corresponding to Motion_mode_ICE/ICE_mode in the model.
Hybrid software mode	Software-mode representation corresponding to Motion_mode_ICE/Hybrid_mode in the model.
Mot_Enable	Binary supervisory output that enables the motor path according to the selected mode.
Gen_Enable	Binary supervisory output that enables the generator path according to the selected mode.

ICE_Enable	Binary supervisory output that enables the engine path according to the selected mode.
ENG_ON	Engine-speed threshold of 800 rpm used to allow transitions out of Start_mode and common Motion_mode_ICE exits.
ENG_OFF	Engine-speed threshold of 790 rpm used to reset ICE_mode or Hybrid_mode back to Start_mode.
SPEED_STOP	Vehicle-speed threshold of 0.5 used for standstill detection.
SPEED_REGEN	Vehicle-speed threshold of 5.0 used for regenerative entry conditions.
SPEED_EV_MAX	Vehicle-speed threshold of 35.0 used for EV eligibility and engine-supported-operation entry.
PDEM_REGEN	Requested-power threshold of -5.0 used to enter regenerative operation.
PDEM_HYB_IN	Requested-power threshold of 50.0 used to leave EV_mode or RegenB_mode for engine-supported operation.
PDEM_HYB_OUT	Requested-power threshold of 40.0 used to allow return from Motion_mode_ICE to EV_mode.
PDEM_HYB_MID	Requested-power threshold of 15.0 used in Start_mode and ICE_mode transition decisions.

PDEM_HYB_LOW	Requested-power threshold of 10.0 used to return from Hybrid_mode to ICE_mode.
SOC_EV_IN	SOC threshold of 0.37 used to enter or return to EV_mode.
SOC_EV_OUT	SOC threshold of 0.35 used to leave EV_mode or RegenB_mode for engine-supported operation.
SOC_MID	SOC threshold of 0.30 used to select Hybrid_mode instead of ICE_mode.
SOC_LOW	SOC threshold of 0.25 used to leave Hybrid_mode for ICE_mode.
SysHLR	High-level system requirement category used for externally visible system behavior.
SwHLR	High-level software requirement category used for mode-logic implementation behavior.
NfHLR	Non-functional requirement category covering timing, resource, stability, and determinism constraints of the software.
VMU	Vehicle Management Unit role represented in this project by the power-split supervisory mode logic and the future software implementation derived from it.